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- `senzhang-resume/internal/2026-05-05-cannondesign-hr-negotiation-brief.{md,pdf}` -
`senzhang.me/internal/2026-05-05-cannondesign-hr-negotiation-brief.{md,pdf}`

Part 1 — Updated Self-Assessment (corrects stale public resume data)

The public resume (`Resume.pdf` , `index.html` , `senzhang.me/src/data/resume.ts`) is out of date in several material ways. Use these **corrected facts** for HR conversations and any updated CV.

Title chain (CURRENT)

- **Computational BIM Developer** — Ennead Architects LLP — 2022 → present
- **BIM Manager** — concurrent role — 2022 → present (NYU Langone, Northwell, FTZ Shanghai, ByteDance, Bilibili)
- **Rhino Doctor** — concurrent role — 2022 → present (Rhino-to-Revit interop, complex-geometry triage)
- Architectural Designer, BIM Specialist — Ennead — 2020–2022
- Designer, BIM Specialist, CG Visualization — Arquitectonica — 2017–2020

Total professional experience post-Master's: ~9 years.

EnneadTab (the IP asset CannonDesign just inherited)

Field	Public resume says	Reality (2026-05)
Tool count	"200+"	450+ (per senzhang-resume/index.html)
Legal status	"3 Copyright applications" / "U.S. Copyright Office"	3 U.S. Copyright registrations (use this exact phrasing — patent ≠ copyright; the HTML loosely says "Patented" but the registrations are copyrights)
Scope	Plugins for Revit/Rhino	Cross-application ecosystem: Rhino, Revit, CAD, Autodesk Desktop Connector, Enscape, InDesign, Photoshop, Excel, Email, Notification, AI/LLM, Computer-Resource Monitoring, TextToSpeech, Sound Effects
Distribution	(not specified)	Auto-publish dev repo → CPython compile → distribution repo → user-side scheduled updater + delayed deletion → silent background updates while app runs
Dependence	"low Grasshopper/Dynamo dependency"	Internally architected to eliminate Grasshopper/Dynamo dependency for office-standard workflows
User base today	Ennead (~160)	About to expand to combined CannonDesign + Ennead (~1,760 staff (Cannon ~1,600 pre-merger + Ennead ~160))

Public profile / speaking (recency leverage)

Year	Venue	Role	Topic
2023	Digital Built Week Americas (Dallas, TX)	Panel Speaker	Promoting computational design to non-programmers
2023	AEC Hackathon (NYC)	Team Honorable Mention	Educational tool linking junior↔senior students
2024	Autodesk University (San Diego, CA)	Session Speaker — "Revit As A Game Engine"	Game-design strategies in Revit (Battleship Hexgrid, Monopoly)
2024	U.S. Copyright Office	3 registrations filed for EnneadTab	—
2026	Advancing Computational Design Conference (Austin, TX)	Speaker — representing Ennead Architects, Design Technology	"The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency" (22 slides + sub-slides; live at 2026-ai-talk.vercel.app ; source repo zsenarchitect/2026-AI-Talk , 153 commits Jan–Mar 2026)

 **Verified:** this is your most recent industry-facing speaking engagement and the freshest piece of leverage. Slot it next to AU 2024 in any handout — it shows your reputation as an Ennead Design Technology speaker is **already public**, in 2026, and now under CannonDesign's banner if it isn't claimed quickly.

Awards / credentials baseline

- Registered Architect, NY (NCARB)
 - Columbia GSAPP — M.S. Advanced Architecture Design
 - RPI — B.Arch (AIA Henry Adams Award, Magna Cum Laude, Faculty Selection Award)
 - Tongji University — Urban Planning major (2010–2012)
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Part 2 — The Project Portfolio CannonDesign Just Inherited

Public resume mentions only EnneadTab. **In reality, the past 12 months of git activity (2025-05 → 2026-05) shows a much larger architectural-tech portfolio that you personally built or led.** Use this section as visual proof of "AVP-scope work" in any negotiation.

Sen-authored commit volume by repo (last 12 months)

Repo	Commits	Window	Type
RenderPolisher	693	2025-11 → 2026-05	Internal AI tool — render post-processing pipeline
NYU-HQ	268	2025-10 → 2025-12	Healthcare BIM/coordination project site
EmployeeData	194	2025-09 → 2026-03	Multi-component employee directory + visualization (live on GitHub Pages)
BimRunner	152	2026-01 → 2026-05	Revit Project Analytics dashboard, 91,027 LOC, 422 files — Next.js 14 + Vercel Postgres
EnneadTabHome	139	2025-12 → 2026-04	Landing page for <code>enneadtab.com</code> — central hub for all EnneadTab apps
2025PerformaceReview	77	2025-08 → 2026-03	Firm-wide employee evaluation engine — Excel ingest + interactive HTML report (live)
AIA-Sustainable-Reporting	53	2026-02 → 2026-04	CRON service for Excel processing + frontend dashboard for AIA 2030 sustainable metrics
EnneadTabLLM	15	2025-12 → 2026-03	Ollama RAG monorepo — Windows server + Vercel proxy + shared TS types
Rosette	5	2026-04	"Architecture of meaning" — design-team/client/government nonsense translator (private EnneadTab product line)
FieldDay	7	2026-03 → 2026-04	Offline-first PWA for field BIM site inspection — phone/tablet/desktop
RemoteKeynote	3	2025-12 → 2026-04	Collaborative Excel editor with real-time editing + auto-save (live demo)
ContentCatalogRunner	5	2026-03 → 2026-04	Internal content cataloguing
ErrorDump	3	2026-03 → 2026-04	Production error aggregation API

Repo	Commits	Window	Type
EnneadTabWiki	(ennead-architects-llp/EnneadTabWiki — last activity 2026-05-02)	private GitHub repo	Public-facing documentation site for the entire EnneadTab ecosystem
2026-AI-Talk	153 (Sen-authored)	2026-01-15 → 2026-03-24	Speaker repo for Advancing Computational Design Conference, Austin TX 2026 — talk title " <i>The Design of Design: Enabling Agile Introduction of New Tools to Scale Design Efficiency</i> " — 22 main slides + sub-slides + presentation notes — React 19 + Vite, deployed at <code>2026-ai-talk.vercel.app</code> , given as Ennead Architects, Design Technology

What this proves to HR

You are not "an architect who happens to code." You are running a **firm-wide internal product organization** that includes:

1. A flagship plugin platform (EnneadTab, 450+ tools)
2. A web analytics dashboard for BIM project health (BimRunner — production-grade, 91k LOC)
3. A firm-wide HR/performance reporting system (2025PerformaceReview — live, used by leadership)
4. A sustainability metrics pipeline tied to AIA 2030 commitment (AIA-Sustainable-Reporting)
5. A field-inspection PWA (FieldDay)
6. An LLM RAG infrastructure (EnneadTabLLM)
7. A render-post-production pipeline (RenderPolisher)
8. A central web hub (EnneadTabHome — `enneadtab.com`)
9. An internal docs system (EnneadTabWiki)

This is the work product of a **Director of Design Technology**, not a Senior Architect. CannonDesign just acquired all of it.

Part 3 — CannonDesign Career Workbook Decode

Level / Title / Corp-title mapping (from the workbook)

Level	Architecture title	Corp Title	Years	Notes
P3	Architect III	Associate	6+	RA preferred
P4	Architect IV / Project Architect IV / Technical Designer IV	Senior Associate	8+	RA required for Architect IV/PA IV
P5	Project Architect V / Technical Designer V	AVP	10+	RA required
P6	Sr. Project Architect I / Sr. Project Designer	VP	12+	RA required
P7	Sr. Project Architect II / Project Technical Director	Principal	15+	RA required
M3	Design Leader I / Sr. Technical Leader / Building Performance Leader	VP	12+	mgmt track
M4	Design Leader II / Associate Technical Director	Principal	15+	mgmt track
M5	Sr. Design Leader / Technical Director	Principal	18-25+	mgmt track

Critical gap

The workbook has no Computational Design / BIM / Design Technology track. But CannonDesign publicly posts these roles on its careers page:

- *Computational Designer* — posted band **\$81,700 – \$91,600** (cannondesign.com/careers/computational-designer, accessed 2026-05-03)
- *Design Technologist* — remote band \$76.8K – \$96K
- *Associate Technical Director*
- *Computational Design Architect* (confirmed: Andy VanMater, St. Louis office)

→ **The workbook you received is incomplete. That is your wedge.**

Part 4 — Market Comp Numbers (memorize the top 3)

Top 3 numbers to have on the tip of your tongue Monday:

1. **NYC Computational Designer average: \$149,444** (Glassdoor, n=19, Dec 2025; 25th \$119,134 / 75th \$189,206 / 90th \$232,790)
2. **NYC Computational Design Architecture 75th percentile: \$199,700** (ZipRecruiter; 25th \$183,200 / 90th \$204,584)
3. **CannonDesign AVP total comp range: \$144K–\$211K** (Glassdoor)

Full table:

Title	NYC base 25th–75th	Median	Source
BIM Manager NYC	\$78–125K	\$100K	Glassdoor
Senior BIM Manager NYC (active posting)	\$150–180K	—	Glassdoor jobs
Computational Designer NYC	\$119–189K	\$149,444 avg	Glassdoor (n=19, Dec 2025)
Computational Design Architecture NYC	\$183–199K	\$191K	ZipRecruiter
Project Architect NYC (8+ yr senior)	—	\$122,882	ERI SalaryExpert (NOT ZipRecruiter — ZR median is lower at \$115.9K)
Senior Architect NYC (10+ yr)	\$140–180K	\$135K	ZipRecruiter
Director of Architecture NYC	\$145–200K	\$154K	Salary.com
Architecture Design Director NYC	\$192–227K	\$209K	Salary.com

Specialty premium: Computational/BIM specialty = **+15% to +35%** over generic Project Architect at the same experience level.

Macro anchors:

- AIA 2025 Compensation & Benefits Report — specialized roles outpaced generalist architect raises
- BLS National Architects 2024: median \$96,690; 90th \$159,800 (NYC ~30–40% above national)
- 2026 NYC inflation: headline 4.0%, housing 5.1%
- AEC technical roles 2026 budget: +4.0% (Zweig)
- US salary increase budgets 2026: 3.5% avg (WTW/SHRM)

→ A flat 4% merit raise just keeps pace with NYC inflation. **Acquisition transitions are the one moment to ask for a discrete level jump (12–25% base + retention).**

Part 5 — Recommended Title Asks (ranked)

Rank	Title	Corp title	Cannon level	Base anchor
🟡 Stable	Senior Associate, Design Technology	Senior Associate	P4	\$150–175K
🟢 Primary target	Computational Design Lead, Associate Vice President	AVP	P5	\$165–185K
🔴 Anchor (lead with this)	Director of Design Technology	Principal/VP	M3	\$200–254K

Tactic: open with the Director ask as anchor; HR rejects; you "compromise" to AVP/Computational Design Lead, which is what you actually wanted.

Industry parity figures (named)

- **Luc Wilson** — Associate Principal / Global Director of Design Technology, KPF (NYC, Columbia GSAPP) — *closest direct comp*
- **James Vandezande** — Senior Principal / Chief Technology Officer, HOK (hok.com/people/leadership/view/james-vandezande/)
- **Jonathan Rabagliati** — Geometry & Computational Design Lead, SOM
- **Andreas Bak** — Global Lead, Computational Design, BIG NYC
- **Bill Carney** — Principal / Design Technology Leader, DLR Group
- **Matt Conway** — Senior Associate / Computational Design Leader, DLR Group (RA — exact rung Sen should target)
- **Brendan Mullins** — Design Computing Discipline Leader, Stantec (CannonDesign's nearest competitor)
- **Zach Trattner** — Senior Associate, Design Technology Leader, Gensler NY (template title to ask for verbatim)
- **Andrew Heumann** — outside-industry parity (NBBJ → WeWork → Hypar)

Part 6 — Monday Game Plan

30-second opening script (memorize)

"Thanks for making time. I want to use this meeting for two things. First, to make sure my mapping into the Cannon ladder reflects what I actually do — I lead computational and BIM development at Ennead, I built and maintain EnneadTab which is now Cannon IP, and I hold three U.S. copyright registrations tied to that platform. Second, I'd like to understand whether I'm being considered for a key-talent retention conversation as part of the integration. I'm not here with an ultimatum — I'm here because the workbook came out three days ago and I'd rather have this conversation now, before the

mapping is finalized, than try to reopen it later. Can we start with how the mapping process works and what the timeline looks like?"

5 must-ask questions

1. **Mapping timeline:** "What's the timeline and process for mapping Ennead staff into the Cannon ladder — when does the mapping become final, and what's the appeal/discussion window?"
2. **The wedge** — *most important question:* "The workbook I received doesn't appear to have a Computational Design or Design-Technology track, but Cannon publicly posts Computational Designer, Design Technologist, and Associate Technical Director roles — can you walk me through how those roles fit into the workbook, and how I'd map into them?"
3. **Band disclosure:** "What's the salary band for [target level — Senior Associate or AVP, Design Technology], and what does Cannon use as its market benchmark — AIA, ZweigWhite, internal?"
4. **Retention list** — *single most important ask:* "Has Cannon identified key-talent retention agreements for the Ennead acquisition, and what's the process for that conversation? I'd like to be considered."
5. **ESOP path:** "Cannon brought Ennead partners into the ownership model. For senior technical staff below partner, is there a path to ESOP participation or an accelerated track, and when would that become available?"

5 likely HR pushbacks + scripted counters

HR says	You say
"The bands are fixed — we can't change them."	"I'm not asking to change the bands. I'm asking which existing band fits the scope of what I do. Can we look at the AVP-Design Technology band against the EnneadTab scope together?"
"We don't have that title at Cannon."	"You actually do — <i>Associate Technical Director</i> and <i>Computational Designer</i> are posted on your careers page. Is the workbook I received the complete version, or is the technology track in a separate document?"
"We'll revisit at integration / in 6 months / at next review."	"I hear that the timing feels early. My concern is that once the mapping is locked, reopening it is much harder — for both of us. Can we agree on a written placeholder now, with a defined 90-day review tied to specific deliverables?"
"Everyone got the same workbook — we're treating people consistently."	"Consistency on process, yes. Consistency on outcome would mean mapping a 9-year registered architect with three U.S. copyright registrations to the same level as a 3-year drafter. What's the consistent <i>process</i> for handling specialist roles?"
"Your current title is X — that's what maps over."	"Title at Ennead reflected Ennead's ladder, which doesn't have a computational track either. The mapping question is: what does my <i>scope</i> map to in Cannon's ladder? Can we pull the AVP and Senior Associate scope definitions and walk through them?"

One-page handout to bring (printed, 2 copies)

SEN ZHANG – MAPPING & RETENTION DISCUSSION – May 5, 2026

1. ROLE SUMMARY

- 9-yr Registered NY Architect (NCARB)
- Computational BIM Developer + BIM Manager + Rhino Doctor (concurrent), 2022–present
- Columbia GSAPP M.S. + RPI B.Arch + Tongji

2. SCOPE OF IMPACT (quantified)

- EnneadTab: 450+ tools, firm-wide deployment (Ennead 160 → Cannon ~1,760 post-merger)
- 3 U.S. Copyright registrations (cite reg. numbers)
- BimRunner: 91k LOC Revit analytics dashboard (lead developer)
- 2025PerformanceReview: firm-wide HR engine (live, used by leadership)
- AIA-Sustainable-Reporting: AIA 2030 metrics pipeline
- FieldDay PWA, RenderPolisher, EnneadTabLLM (RAG), Rosette, EnneadTabHome
- Speaking: Autodesk University 2024, Digital Built Week 2023, [Austin 2026]
- Healthcare BIM Manager: NYU Langone, Northwell
- China commercial BIM Manager: FTZ Shanghai, ByteDance, Bilibili

3. PROPOSED CANNON MAPPING

Target: Associate Vice President, Design Technology
(or: Computational Design Lead, Senior Associate)

Evidence Cannon uses these roles: posted "Computational Designer",
"Associate Technical Director", "Computational Design Architect"
(Andy VanMater, St. Louis office)

4. COMP BENCHMARKS

- AIA 2025 Compensation Report (specialty premium documented)
- Cannon's posted bands: \$81.7K (Computational Designer) → \$211K (AVP)
- Industry comps: Luc Wilson (KPF), James Vandezande (HOK),
Matt Conway (DLR), Zach Trattner (Gensler)

5. RETENTION ASK

Request consideration for retention agreement; 24-mo term tied
to EnneadTab / BimRunner integration milestones

6. WHAT I'M NOT ASKING FOR

- x Not asking to change bands
- x Not asking partner-track equity yet
- x Not threatening to leave

7. NEXT STEP

Written response within 14 days · Comp committee review before
mapping locks

"Talent" terminology — Chinese ↔ English term ladder ⚠ critical

The Chinese word 人才 translates loosely as "talent," but the English **"talent"** is too soft for HR — it sounds like self-praise and triggers nothing. Use the English ladder below to escalate:

中文	English usage	What it triggers in HR
人才	"talent"	Generic praise — triggers nothing
关键人才	"key talent"	Triggers retention-list evaluation process
关键技术人才	"key technical talent"	Same + triggers "specialist exception" process
核心人才	"key personnel" / "critical-role employee"	M&A legal language — triggers retention agreement process
(最强)	"key talent retention agreement"	Directly maps to the same contract framework Cannon is already using with Ennead partners — just adds you to the list

Use this exact phrasing on Monday (already baked into question #4):

*"Has Cannon identified **key-talent retention agreements** for the Ennead acquisition, and what's the process for that conversation? I'd like to be considered."*

Three things this sentence does at once:

1. Acknowledges Cannon is already doing this (the partner ownership conversion is public news)
2. Doesn't directly say "I am key talent" — lets HR judge (Voss-style calibrated question)
3. Routes you into an existing budget pool instead of forcing HR to break the base-salary band

Optional escalation — "single point of failure" framing

If you want to upgrade the talent frame one more notch, borrow IT/risk vocabulary — to HR this sounds objective rather than self-promoting:

"EnneadTab, BimRunner, and EnneadTabLLM are single-point-maintenance products. If the integration mismaps me, Cannon faces a 90-day continuity risk on three production systems. I'd like to be included in the key-talent retention conversation — not as self-promotion, but to put product continuity into the contract."

This translates "I am talent" into **risk-management language** — HR almost has to escalate this to a Practice Leader, which lengthens your lever.

Mandarin-language colleagues (Ennead Asia partners, private)

In Mandarin private conversations with Ennead's China-side partners (e.g., 安栋林 Andrew Bartle's circle), you can be direct:

"周一我会跟 HR 谈 mapping，希望被认作**关键技术人才**，进入收购整合的 retention 名单。EnneadTab 系列产品离开我维护不下去，这点想请你心里有数。"

This primes them as informal allies before the formal HR conversation locks.

IP framing (memorize)

"EnneadTab is firm IP — that's clear. What I want to talk about is making sure the integration captures its full value. Cannon now owns 450+ tools and 3 registered works; I'd like to discuss what role lets me extend that platform across Cannon's 1,600+ staff rather than just Ennead's 160."

This reframes you from "cost center asking for raise" to "strategic asset asking for scope" — same ask, completely different valence.

Walk-away tiers (private, not voiced)

-  **Accept-and-document:** Mapping at Senior Associate or above + written retention conversation scheduled within 30 days + salary band disclosed
-  **Yellow** (keep talking): Vague "we'll revisit" without date, no retention list mention
-  **Red flag** (start interviewing externally, quietly): Mapping below current Ennead level, no retention conversation, written denial of band access → reach out to NBBJ, HOK, SOM, Perkins+Will, KPF
-  **Hard walk:** Only after a competing written offer + final no from Cannon — never on emotion. **Do NOT walk Monday.**

Post-meeting checklist

Window	Action
24 hours	Send recap email to HR + Ennead reporting partner — bullet what was discussed, what was agreed, what's open
7 days	Quietly update LinkedIn + résumé (not to leave — to know your market number cold)
14 days	Follow-up email if no written response
30 days	If no retention conversation has materialized, escalate to Ennead managing partner
45 days	If still nothing, pivot — either accept and start "earn the next level" play, or start external interviews
60 days	Mapping likely locks

Part 7 — Source Index

Primary documents

- [~/Downloads/CannonDesign Career Workbooks.eml](#) (2026-04-30 from Kevin McClurkan, Ennead Partner)
- [Career Workbook Architecture and Interiors 042026.pdf](#) (40 pp, internal, dated April–May 2025)
- [senzhang-resume/ repo HEAD b51e2e0](#) (2026-04-30)
- [senzhang.me/src/data/resume.ts HEAD 4d333ed](#) (2026-03-21 — older, do not use as source)

Acquisition press

- [archpaper.com/2026/04/cannondesign-acquires-ennead-architects/](#)
- [architectmagazine.com](#) — "One of Architecture's Most Strategic Mergers Just Dropped"
- [cannondesign.com/news/ennead-architects-joining-cannondesign-deepen-designs-impact-society](#)
- [bisnow.com](#) — CannonDesign Acquires Ennead Architects Amid Rapid Growth Spurt
- [architecturalrecord.com/articles/18150-cannondesign-acquires-ennead-architects](#)
- [bloomberg.com/news/articles/2026-04-21/cannondesign-acquires-ennead-architects-in-major-design-firm-merger](#)

Cannon role evidence (the wedge)

- [cannondesign.com/careers/computational-designer](#)
- [cannondesign.com/careers/design-technologist](#)
- [cannondesign.com/careers/associate-technical-director](#)

Comp data

- [glassdoor.com/Salaries/new-york-city-ny-computational-designer-salary](#)
- [ziprecruiter.com/Salaries/Computational-Design-Architecture-Salary-in-New-York-City,NY](#)
- [glassdoor.com/Salary/CannonDesign-Salaries-E30940](#)
- [aia.org/about-aia/press/aia-compensation-benefits-report-2025-now-available](#)
- [bls.gov/ooh/architecture-and-engineering/architects.htm](#)
- [bespokecareers.com/salary-guide/new-york/architecture/](#)
- [shrm.org](#) — Employer Salary Increase Predictions 2026
- [zweiggroup.com/blogs/news/aec-salary-trends-2026-compensation-data](#)

M&A negotiation references

- [mckinsey.com](#) — Talent retention and selection in M&A
- [pearlmeyer.com](#) — Sizing Up Retention and Transaction Bonus Pools
- [capartners.com](#) — Compensation Implications of Mergers and Acquisitions
- [inc.com](#) — Being Acquired? Here's How You Negotiate Your Retention
- Voss, *Never Split the Difference* — Black Swan Group
- Fisher / Ury / Patton, *Getting to Yes* — BATNA + objective criteria

IP / copyright

- aia.org/resource-center/understanding-copyright-protection-architects
 - axley.com/publication_article/copyright-ownership-of-creative-works-made-for-hire/
 - venable.com — Understanding the Work Made for Hire Doctrine
 - 17 U.S.C. § 101 (Work-for-hire doctrine — code created by salaried employee within scope is owned by employer; author credit ≠ ownership)
-

Part 8 — Design Technology Vision: Why This Conversation Matters Beyond Comp

8.1 Where AEC design tech is going (2026-2030)

The next four years of architecture practice will be defined by one shift: design intent stops being expressed primarily as drawings and starts being expressed as code, constraints, and prompts that machines can act on. This is no longer speculative. It is happening on three concurrent tracks, each with named, funded products in the market today.

Track 1 — Generative early-stage design. Autodesk's Forma platform, the renamed and expanded successor to Spacemaker, demonstrated "Neural CAD" at Autodesk University 2025: set a building type and structural material, and the system produces a complete interior layout with unit counts in seconds, trained on Autodesk's own architectural corpus rather than a general model. Forma Building Design enters general availability in 2026 and targets schematic phases (LOD 200–300) directly inside Revit through the new Forma Connect Client (bimpure.com/blog/au2025; autodesk.com/products/forma/overview).

Bentley's parallel move, **OpenSite+**, is the first production civil engineering tool driven by generative AI — Bentley publicly claims projects delivered up to 10× faster, with Bentley Copilot rolling into OpenRoads and OpenRail in early 2026 and ProjectWise AI search going GA in 2026 (bentley.com/news; aecmag.com/ai/bentley-systems-shapes-its-ai-future).

Track 2 — Automation of the documentation and engineering middle. Augmenta's Construction Platform routed the electrical system at Mt. Hope Elementary in August 2025 — the first US building with an AI-designed electrical raceway network — collapsing six-to-eight weeks of BIM modeling per phase into days, with auto-plumbing slated for 2026 (Globe Newswire 2025-08-05; Globe Newswire 2025-09-11 Augmenta Agentic Design Environment). On the construction side, **Bluebeam Max** (launching globally in early 2026) bundles a Revu + Anthropic Claude integration plus AI-REVIEW / AI-MATCH from Bluebeam's Firmus acquisition (press.bluebeam.com Bluebeam Unbound 2025). **Autodesk Assistant**, embedded in Construction Cloud, lets project teams ask plain-language questions of their own model and contract data (ENR: Autodesk folds Construction Cloud into Forma).

Track 3 — Architecture-as-code platforms, rendering, and feasibility. **Hypar** (Andrew Heumann, ex-NBBJ / WeWork) ships generative space-planning algorithms benchmarked against 13,000 real offices, with text-to-BIM mapping natural language to parameter sets (aecmag.com Hypar 2.0; hypar.io). **Higharc** has fully automated single-family home documentation: every plan is a parametric program that generates drawings, takeoffs, and showroom assets — Higharc AI launched broader beta in early 2025 (higharc.com

newsroom). **TestFit** owns developer feasibility (testfit.io 2025 Year in Review). **Maket.ai** generates compliant residential floor plans from constraints (maket.ai). **Veras** (now part of Chaos alongside V-Ray and Enscape) and **LookX / Arko AI** put diffusion-based rendering directly on top of Revit, Rhino, SketchUp, Archicad, Vectorworks, and Forma 3D models (blog.chaos.com AI tools for architects 2026). **Architectures** does the same for early massing.

The research scaffolding around all of this — ACADIA, Smart Geometry, AIA TAP — has not stood still either. ACADIA 2025 in Miami (theme: *Computing for Resilience*) was explicit that climate, equity, and economic uncertainty are now first-class computational design problems, not aesthetic ones (archpaper.com 2025/10 ACADIA 2025; 2025.acadia.org). The TAP Knowledge Community at AIA25 has formally folded technology and sustainability into one symposium (AIA Community Hub).

The throughline: by 2028 the question for a design firm will not be "do you use AI" but "do you own any of the IP that produces your buildings, or are you renting it from Autodesk, Bentley, and a half-dozen startups?"

8.2 Ennead's design-tech lineage — what was, what's gap

Ennead has a real, public computational identity, but it has lived more in *project artifacts* than in *productized infrastructure*. The Shanghai Astronomy Museum is the canonical example: a building with no straight lines or right angles, geometry derived from the three-body problem, a façade and circulation system generated parametrically — covered as a parametric exemplar in the trade press (parametric-architecture.com; archdaily.com; dezeen.com 2021/08/09). Stanford's Bing Concert Hall (Richard Olcott, 2013) integrated acoustics, video, and form into one organic envelope, a reference point for architecture-as-instrument (designboom.com Ennead Bing Concert Hall). **Ennead Lab** (e-Lab), led by Director **Amy Mielke**, runs the public R&D arm — Starter Home 3D-printing, WALTER smart water infrastructure, the Bird-Friendly Glass Lab, refugee-community design toolkits with UNHCR (e-lab.ennead.com) — and recently exhibited *Light Speed*, a flexible metal + UHPC composite panel system, at AMAD 2025 (ennead.com news on Ennead Design Technology in Parametric Architecture).

What is publicly *claimed* is striking. Ennead's own marketing language now states the firm is "advancing the boundaries of computational and parametric design, exploring the implications of quantum computing for the built environment and shaping how AI is transforming architecture" — language echoed verbatim in the CannonDesign acquisition press (cannondesign.com news Ennead joining CannonDesign).

What is *missing*, observably, is a named in-house computational platform with a public identity — no Ennead-branded equivalent of SOM's design-technology group, NBBJ's NBBJ Digital, HOK's HOK Forward, Foster + Partners' Applied Research and Development team, or Perkins&Will's AREA Research. e-Lab's published roster is heavily research and advocacy; the firm has no publicly named director of computational design, no published platform in the SOM / Foster / NBBJ mode, and no cited Autodesk University, ACADIA, or Smart Geometry presence at the leadership level. **The gap between Ennead's stated computational ambition and its productized, named, internally-staffed design-tech infrastructure is the single largest delta the merger inherits.** That delta is the strategic opening.

8.3 The Cannon + Ennead design-tech thesis

CannonDesign and Ennead are two firms that, on paper, look more different than they are. Cannon is 1,600+ people across 20 offices, employee-owned, healthcare- and higher-ed-anchored, with a publicly trademarked **Living-Centered Design** methodology that names interdependencies between people, business, community, and environment as the design substrate (cannondesign.com/about/living-centered-design; cannondesign.com/about/). Ennead is 160 people across NYC, LA, and Shanghai, design-forward, civic and cultural, with a global commercial book in Asia (Architectural Record; Architect Magazine; Archpaper).

The merger language is unusually explicit on *why* technology matters to the combined firm. From the joint press release: *"the goal is not just to improve technologies entering the design marketplace, but to also invent them"* — naming computational design, quantum computing, and AI applications by name. CEO Brad Lukanic, in Bisnow, frames the deal verbatim: *"We have found that some firms are thinking growth for growth's sake. We're thinking growth for differentiation."* Architectural Record adds: *"We want to grow a 100-year business that is for our next generation of people."*

The strategic implication is sharper than it looks. Cannon brings the **scale, healthcare data sets, and ESOP structure** — which means anyone who builds internal tooling builds equity that flows back to employee-owners, not to a private-equity captable. Ennead brings the **design-forward institutional brand and the Shanghai book** — which means tools built here can be marketed externally as design-led, not engineering-led. Healthcare specifically is the sharpest AI-in-AEC vertical of the next five years: clinical workflow simulation, MEP density, infection-control airflow modeling, and code-compliance automation are exactly the problems where Augmenta-class agentic tools and LLM-driven plan checking pay back fastest. A combined firm with Ennead's design polish, Cannon's healthcare data, and a real internal design-tech platform would be one of three or four shops in the world positioned to *productize* — not just consume — the next wave.

8.4 The AI-in-AEC moment, named in concrete products

The macro signal is unambiguous and compressed into the last twelve months:

- **Turner Construction**, the largest US contractor by revenue, signed a two-year "wall-to-wall, floor-to-ceiling" deal with OpenAI — every employee gets ChatGPT Enterprise, plus a hackathon that produced 100+ custom agents for contract review, safety, and jobsite ops (constructiondive.com; constructionbriefing.com).
- **AECOM** acquired Norwegian AI startup **Consigli** for **\$390M** in November 2025 — the largest AI acquisition in AEC engineering to date (verdantix.com; moginlawllp.com). The trade analysts explicitly name **WSP, Jacobs, Arup, and Bechtel** as firms that "must respond quickly."
- **Autodesk** reorganized: Construction Cloud is folding into Forma, and Forma Building Design enters GA in 2026 with neural-CAD layout generation (ENR).
- **Bentley** committed publicly to AI-first infrastructure, with OpenSite+, Bentley Copilot in OpenRoads/OpenRail by early 2026, ProjectWise AI search GA 2026 (bentley.com press).
- **Dynamo** added native agentic workflows and MCP (Model Context Protocol) support for LLM-tool calling inside Revit graphs (dynamobim.org; forum.dynamobim.com Agentic workflows). Third-party Revit MCP servers are shipping (archilabs.ai/posts/revit-model-context-protocol).
- **Bluebeam Max** (Anthropic Claude inside Revu) ships globally early 2026 (architosh.com).

- A 2024–25 Chaos / Architizer survey of 1,227 architects found **46% already using AI tools at work, with another 23% planning adoption** within 12 months (blog.chaos.com).

Translation for an HR audience: every direct competitor of Cannon-Ennead either bought or built its AI capability in the past 18 months. The firms that did not are losing senior technologists to those that did.

8.5 Why this matters for Sen's role specifically (cover-page pitch, ~250 words)

I am not asking for a title and a raise because the M&A timing is convenient. I am asking because the combined firm has publicly committed — in its own acquisition press — to *invent* design technology, not just consume it. That promise has to be staffed by someone who has already been doing exactly that, in production, inside Ennead.

For the past several years I have built and shipped **EnneadTab**, the firm's internal Revit and Rhino automation platform, used daily across NYC, LA, and Shanghai studios. I built **BimRunner**, our automated model-audit and QA pipeline. I built **EnneadTabLLM**, an Ollama-based local RAG system that lets staff query our standards, specs, and prior projects without sending data to a vendor — solving the same problem CannonDesign healthcare clients will need solved under HIPAA. I built our **AIA Sustainable Reporting** automation, the kind of code-as-compliance work that Bluebeam Max and Bentley Copilot are now charging enterprise prices for. None of this is theoretical. All of it is in production. All of it was built without a "Director of Computational Design" title because Ennead never created one.

Cannon-Ennead's stated thesis is that the combined firm will invent design technology in healthcare, civic, and cultural work, leveraging Living-Centered Design and Ennead's institutional brand. I am the person who has already been quietly doing that work — bridging Ennead's design culture with Cannon's healthcare scale, the firm's Asia book with US compliance, and Autodesk's tooling with our own IP. The title and compensation conversation is not about the past. It is about whether the firm wants the person who builds its 2026–2030 design-tech platform to do that work *inside* the building, or watch him build it somewhere else.

End of internal brief. Update this document after the Monday meeting with what HR actually said — that becomes the foundation for the 14-day follow-up and any subsequent escalation.

Appendix A — Fact-Check Verification (completed 2026-05-03)

Every claim used in this brief was independently verified against live sources on 2026-05-03. Below is the full audit table — bring this if HR challenges any number.

ID	Claim	Status	Source	Note
A1	Acquisition announced April 21, 2026	✓ CONFIRMED	archpaper.com/2026/04/cannondesign-acquires-ennead-architects/; Bisnow timestamped Apr 21, 2026 11:33 AM ET	—
A2	Ennead = 160-person practice, NYC/LA/Shanghai	✓ CONFIRMED	cannondesign.com/news/ennead-architects-joining-cannondesign — exact quote	—
A3	7th acquisition since 2024, largest	✓ CONFIRMED	Bisnow + AN	"CannonDesign's seventh acquisition since the beginning of 2024, and Ennead Architects' 160-person practice is CannonDesign's largest acquisition to date"
A4	Employee-owned; CEO Lukanic	⚠ PARTIAL	prnewswire.com Lukanic announcement	CEO effective July 28, 2016 (announced 2015). Sources say "employee-owned" — do not claim "100% ESOP" without a specific source
A5	Operates as "Ennead, a CannonDesign company"	✓ CONFIRMED	CannonDesign press release	—
A6	Cannon 2026 revenue ~\$228.6M; ~1,600 staff pre-merger	✓ CONFIRMED	rocketreach.co + Bisnow	Pre-merger headcount ~ 1,600 (NOT ~1,000); ~1,760 post-merger
B1	Computational Designer band	● CORRECTED	cannondesign.com/careers/computational-designer	Posted band is \$81,700–\$91,600 (NOT \$83,600–\$104,500)
B2	Design Technologist band \$76,800–\$96,000	✓ CONFIRMED	cannondesign.com/careers/design-technologist	A second senior posting shows \$83,600–\$104,500
B3	Architect III NYC \$87,000–\$113,365	✓ CONFIRMED	cannondesign.com/careers/architect-iii-3	—
B4	Associate Technical Director	✓ CONFIRMED	cannondesign.com/careers/associate-technical-director	20 yrs exp, RA license required

ID	Claim	Status	Source	Note
	role exists			
B5	Andy VanMater = Computational Design Architect, CannonDesign St. Louis	✓ CONFIRMED	linkedin.com/in/andyvanmater/	—
C1	Glassdoor NYC Computational Designer	● CORRECTED	glassdoor.com/Salaries/new-york-city-ny-united-states-computational-designer-salary	Average \$149,444 (n=19, Dec 2025) — not \$144,911
C2	ZR NYC Computational Design Architecture 75th \$199,700	✓ CONFIRMED	ziprecruiter.com/Salaries/Computational-Design-Architecture-Salary-in-New-York-City,NY	—
C3	Glassdoor NYC BIM Manager \$78,584–\$124,759	✓ CONFIRMED	glassdoor BIM Manager NYC	n=57, Apr 2026
C4	Bespoke NYC BIM Manager \$95K–\$150K	✓ CONFIRMED	bespokecareers.com/salary-guide/new-york/architecture/	—
C5	Project Architect 8+ yr median \$122,882	● CORRECTED SOURCE	salaryexpert.com — ERI , NOT ZipRecruiter	ZipRecruiter NYC PA median is \$115.9K
C6	Salary.com NYC Director of Architecture	✓ CONFIRMED	salary.com	Avg \$154,221; 25–75 \$143K–\$164K; 10–90 \$133K–\$172K
C7	Glassdoor NYC VP Architecture \$274,027	✓ CONFIRMED	glassdoor VP Architecture NYC	Small sample (n=7) — flag if used as primary anchor
D1	CannonDesign comp rating	● CORRECTED	glassdoor Reviews E30940	3.5/5 (up 7% YoY) — NOT 3.1/5
D2	CannonDesign overall 4.0/5	✓ CONFIRMED	same	n=329 reviews
D3	CannonDesign Architect \$98K avg, \$102K–\$188K	✓ CONFIRMED w/ caveat	glassdoor CannonDesign Architect	This is the 25th–75th percentile band, all-locations , NOT NYC. Reframe accordingly
D4	"Promotion structure is limiting" review	✓ CONFIRMED (paraphrase)	glassdoor reviews	Substance verified; if quoting verbatim, refresh phrasing

ID	Claim	Status	Source	Note
				from current review text
E1	AIA 2025 Comp Report 13,227 positions / 817 firms	✓ CONFIRMED (headline only)	aia.org/resource-center/aia-compensation-benefits-report	Full report paywalled
E2	BLS 17-1011: median \$96,690 / 90th \$159,800 / 123,600 jobs	✓ CONFIRMED	bls.gov/ooh/architecture-and-engineering/architects.htm	—
E3	NYC March 2026 CPI 4.0% / 3.3% / 5.1%	✓ CONFIRMED	bls.gov/regions/northeast/news-release/consumerpriceindex_newyork.htm	—
E4	3.5% US 2026 salary budget	✓ CONFIRMED	worldatwork.org WTW survey, n=1,569 US orgs	Attribute to WTW (corroborated by Mercer/PayScale) , not SHRM
E5	Zweig 2026 AEC technical roles +4.0%	✗ NOT FOUND	zweiggroup.com — no public 4.0% headline	Use category specifics instead (Architectural Designer +11%, \$70K→\$78K)
F1	Luc Wilson — Global Director of Design Technology, Associate Principal, KPF; Columbia GSAPP adjunct	✓ CONFIRMED	kpf.com/people/luc-wilson	—
F2	James Vandezande — HOK	● CORRECTED	hok.com/people/leadership/view/james-vandezande/	Senior Principal / Chief Technology Officer (NOT Design Technology Manager) — <i>upgrade</i> : this is a stronger comp
F3	Jonathan Rabagliati — Geometry & Computational Design Lead, SOM	✓ CONFIRMED	LinkedIn	—

ID	Claim	Status	Source	Note
F4	Andreas Bak — Global Lead Computational Design, BIG	✓ CONFIRMED	LinkedIn	—
F5	Matt Conway — Sr Associate / Computational Design Leader, DLR Group, RA	✓ CONFIRMED	LinkedIn	—
F6	Bill Carney — Principal / Design Technology Leader, DLR Group	✓ CONFIRMED	LinkedIn	—
F7	Brendan Mullins — Design Computing Discipline Leader, Stantec	✓ CONFIRMED	stantec.com/en/people/m/mullins-brendan	Title: "Design Computing Discipline Lead, Principal, Architect"
F8	Zach Trattner — Sr Associate / Design Technology Leader, Gensler NY	✓ CONFIRMED	LinkedIn	—
F9	Andrew Heumann — formerly NBBJ/WeWork → Hypar	✓ CONFIRMED	LinkedIn	Path: NBBJ → Woods Bagot → WeWork → Hypar
G1	Architects Newspaper coverage	✓ CONFIRMED	archpaper.com 2026/04/cannondesign-acquires-ennead-architects/	—
G2	Architectural Record coverage	✓ CONFIRMED (paywalled)	architecturalrecord.com/articles/18150	Headline only — body paywalled
G3	Bisnow uses "rapid growth spurt"	✓ CONFIRMED	bisnow.com NY	Headline match
G4	Lukanic quote	● CORRECTED	Bisnow	Verbatim: "We have found that some firms are thinking growth for growth's sake. We're thinking growth for

ID	Claim	Status	Source	Note
				<i>differentiation.</i> — NOT "growth for differentiation, not growth alone"

Top-5 corrections applied to brief

1.  Computational Designer band fixed: \$83.6K → **\$81.7K**
2.  Glassdoor comp rating fixed: 3.1/5 → **3.5/5**
3.  CannonDesign headcount fixed: 1,000+ → **~1,600 pre-merger / ~1,760 post-merger**
4.  Lukanic quote: paraphrase → **verbatim** Bisnow text
5.  Project Architect \$122,882 source: ZipRecruiter → **ERI SalaryExpert**

Secondary corrections also applied

- Glassdoor Computational Designer NYC: \$144,911 → **\$149,444 average** (improved data)
- Vandezande title: "Design Technology Manager" → **"Chief Technology Officer at HOK"** (stronger industry-parity figure)
- Ennead conference talk: placeholder → **Advancing Computational Design Conference, Austin TX 2026** ("The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency", source repo `zsenarchitect/2026-AI-Talk` , live at `2026-ai-talk.vercel.app`)

End of internal brief. Update this document after the Monday meeting with what HR actually said — that becomes the foundation for the 14-day follow-up and any subsequent escalation.